

Nevena Gligić

PhD candidate, Department of Statistics and Data Science

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EDUCATION

The University of Texas at Austin

PhD Statistics

Academic advisor: Dr. Arya Farahi

Austin, TX

Aug 2022 – Jun 2027

University College London (UCL)

MSc Computational Finance (Final grade: First Class)

London, UK

Sep 2020 – Sep 2021

The University of Manchester

BSc (Hons.) Mathematics (Final grade: Upper Second Class (2:1))

Manchester, UK

Sep 2017 – May 2020

RESEARCH PROJECTS

Shrinkage-Gated Uncertainty Estimation: Combining Prior Beliefs and Density Estimates (*ongoing*)

- RQ: How to make an AI system more reliable under distributional shifts?
- Impact: fraud detection, medical diagnosis, autonomous driving, disaster response management
- Working with: normalizing flows (MAF), evidential deep learning (EDL)

BPSD: Unsupervised Bayesian Probabilistic Signal Detection in Noisy Environments (*ongoing*)

- RQ: How to detect signal in low SNR regime without labels and assumptions on the signal distribution?
- Impact: gravitational waves detection, galaxies detection, medical imaging, radar systems
- Working with: normalizing flows (MAF), CNN, GMM, AE

AI-Driven Utility Monitoring and Anomaly Detection for City-Operated Buildings (*ongoing*)

- RQ: How to detect extreme energy consumption?
- Impact: improved operational efficiency, reduced waste, lower emissions, cut costs
- Working with: Chronos Bolt, transformers, ARIMA

ACHIEVEMENTS AND AWARDS

- *College of Natural Sciences PhD Fellowship (2022-2027)*
- *The School of Mathematics Entrance Scholarship (2017)*

EXPERIENCE

The University of Texas at Austin

Graduate Research Assistant (Dr. Arya Farahi)

Austin, TX (USA)

May 2023 - August 2023, May 2024 - Present

- (Dr. Arya Farahi) Collaboration with the City of Austin. See “AI-Driven Utility Monitoring ...” project.
- (Dr. Mingyuan Zhou) Worked on understanding diffusion probabilistic models.

The University of Texas at Austin

Teaching Assistant

Austin, TX (USA)

Aug 2022 – May 2024

- Lead office hours and provided one-on-one support for students in different introductory statistics classes.
- Provided simple explanations of complex statistical concepts to students new to R programming.
- Graded assignments and provided constructive feedback to help students improve.

University College London, Santander Bank UK

Summer Research Intern

London, United Kingdom

Jun 2021 – Sep 2021

- Project title: Operational Risk Bayes Neural Network
- Research question: What change in operational losses values caused a change in operational risk measures?
- Applied neural networks, naive Bayes classifier, and multinomial logistic regression. Used Python and R.

SKILLS

- **Programming:** Python, R, MATLAB, LaTeX
- **Languages:** Croatian (Native), English (Fluent), German (Intermediate), Italian (Beginner)